



Product Health Diagnostic & KPI Analytics Platform - Business Case Study

Product Performance · Funnel & User Retention Analysis

1. The Problem

The product was getting users, generating orders, and bringing in revenue, but it was not clear **where the product was performing well and where users were getting stuck**.

The product team wanted a simple way to answer questions like:

- Are users successfully moving through the ordering journey?
- Where are users dropping off?
- Are users coming back after their first interaction?
- Which acquisition channels bring better-quality users?
- Which features are actually being used?
- Are product experiments improving conversion?

My goal was to bring these different signals together and create a single view of overall product health.

2. What I Built

I built a **Product Health Diagnostic & KPI Analytics Platform** using:

- **SQL** for funnel, retention, churn, acquisition and experiment analysis
- **Python** for exploratory data analysis
- **Power BI** for the final dashboard and KPI reporting

The dashboard was divided into four main areas:

1. Executive Product Health
2. Product Funnel & Conversion
3. User Retention & Churn
4. A/B Testing & Experiments

The SQL was used to calculate metrics such as ordering users, funnel conversion, drop-offs, retention, churn, feature adoption and experiment performance.

3. First Finding — Overall Product Health

The overall **Product Health Score** was 70.9/100.

At first glance, the product was not performing badly, but a few metrics stood out.

Payment success was 95%, which was above the 90% goal. So payment processing did not look like the main problem.

On the other hand:

- **Order success: 72%** vs 90% goal
- **Churn: 37%** vs 15% goal
- **Feature adoption: 61%** vs 50% goal

This helped narrow the problem down. The bigger opportunities were around **successful order completion and user retention**, rather than basic payment success or overall feature usage.

4. Funnel Analysis — Where Are Users Dropping?

I then looked at the user journey from session to payment.

The main conversion rates were:

- Session → Search: **78.67%**
- Search → Restaurant View: **100%**
- Restaurant View → Cart: **74.58%**
- Cart → Checkout: **88.64%**
- Checkout → Payment: **91.03%**

The SQL also showed that **30 sessions were lost between restaurant view and cart**, while only 10 were lost between cart and checkout and 7 between checkout and payment.

What I found

The bigger drop-off happens **before checkout**.

Once a user adds something to the cart, they are generally more likely to continue.

So instead of immediately focusing on payment or checkout, I would first investigate the **search, restaurant selection and menu-to-cart experience**.

5. Acquisition Channel Analysis

I also compared conversion across acquisition channels.

The overall funnel conversion was:

Acquisition Channel	Conversion
Influencer	60.00%
Google Ads	56.25%
Instagram Ads	46.34%
Referral	41.94%
Organic	41.67%

The SQL analysis showed a clear difference between channels.

This made me look beyond just the number of users coming from each channel.

For example, a channel bringing fewer users may still be valuable if those users convert and continue using the product.

6. Retention & Churn

The product had **63% retention and 37% churn**.

The segment analysis showed that retention was much stronger among highly engaged users.

User Segment	Retention	Churn
Power User	74%	26%
Active	71%	29%

User Segment	Retention	Churn
New	56%	44%
At Risk	55%	45%
Churned	47%	53%

The SQL analysis also identified higher churn among New and At Risk users compared with Active and Power Users.

The biggest question is not just "**Why are users churning?**"

| **What are retained users doing differently during their first few sessions?**

That could help identify opportunities to improve onboarding and early engagement.

7. Acquisition & Retention

Retention also varied by acquisition source.

- Google Ads — **77.78%**
- Referral — **70.00%**
- Organic — **65.22%**
- Influencer — **60.00%**
- Instagram Ads — **48.28%**

The difference suggests that acquisition quality should be evaluated using more than just acquisition volume.

I would look at **conversion, retention and revenue together** when comparing channels.

8. Feature Adoption

Overall feature adoption was **61%**, but individual features were used differently.

- AI Recommendations — **22%**
- Coupons — **17%**
- Live Tracking — **17%**
- Scheduled Delivery — **17%**
- Reorder — **9%**

The SQL calculates adoption based on the percentage of users using each feature.

AI Recommendations had the highest adoption, while Reorder had the lowest.

For lower-adoption features, I would investigate whether the problem is **discoverability, usability, or simply that the feature is not relevant to enough users.**

9. A/B Testing

I also included experiment analysis to understand whether product changes were improving conversion.

For the selected experiment:

- Variant A: **56.25%**
- Variant B: **53.85%**
- Relative difference: **4.27% for B**

The dashboard marked the result as **not statistically significant.**

So I would not make a product decision based only on this observed difference. The experiment would need more data or further testing before treating the result as conclusive.

Across the other experiments, the SQL showed both positive and negative observed differences between variants.

10. What I Found Overall

After looking at the different parts of the product, the main issues were:

1. Order completion

Order success is only 72%, despite payment success being strong.

2. Early funnel drop-off

A noticeable number of users are lost before reaching the cart.

3. User retention

A **37% churn rate** means there is a sizeable group of users who do not continue engaging.

4. Acquisition differences

Conversion and retention vary considerably across acquisition channels.

5. Uneven feature adoption

Some features are being used much more than others.

11. Recommendations

Based on the analysis, I would focus on:

Improve the discovery experience

Look into search, restaurant selection and the journey from restaurant view to cart.

Investigate unsuccessful orders

Break the 28% unsuccessful-order gap down by failure reason, device, user segment and other relevant dimensions.

Improve new-user activation

Compare the early behavior of retained and churned users to identify where new users are losing interest.

Measure acquisition quality

Track conversion and retention along with acquisition volume.

Continue testing product changes

Use A/B testing, but make rollout decisions only after checking statistical significance and having enough data.

12. How I Would Measure Success

I would continue tracking:

- Product Health Score
- Order Success Rate
- Churn Rate
- Retention Rate
- Funnel Conversion
- Feature Adoption
- Acquisition-channel conversion and retention

- A/B test performance

The main goal would be to **improve order completion and retention while reducing unnecessary funnel drop-off.**